

Practical Laboratory 1

Two VM Network Commissioning and Connectivity Verification

Student identification

Full name

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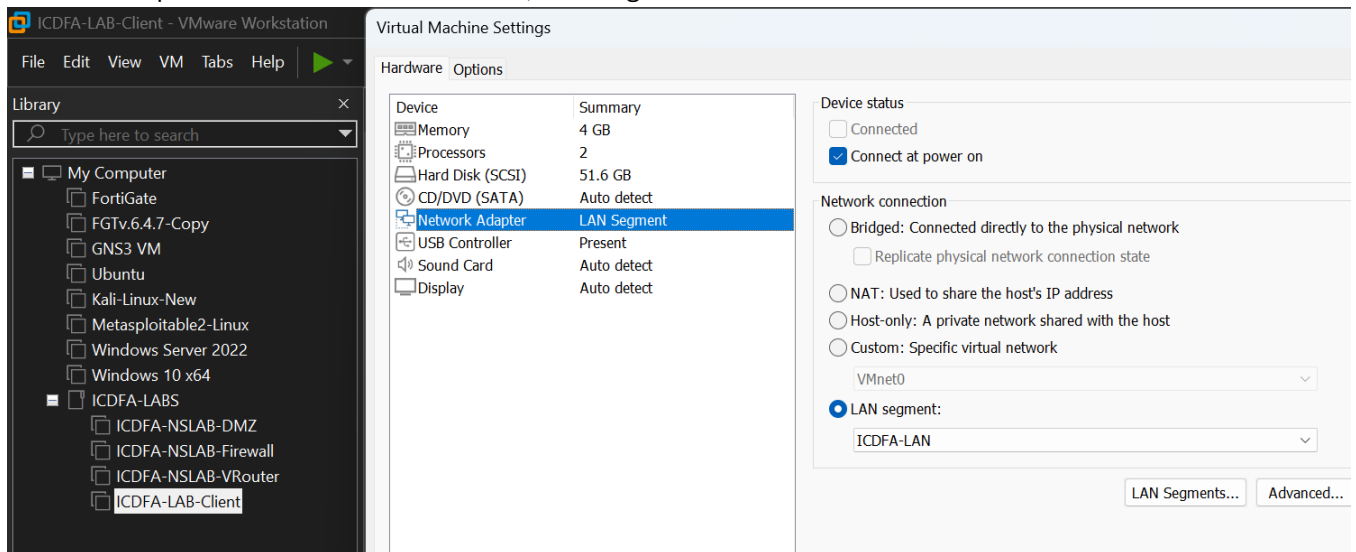
Registration number

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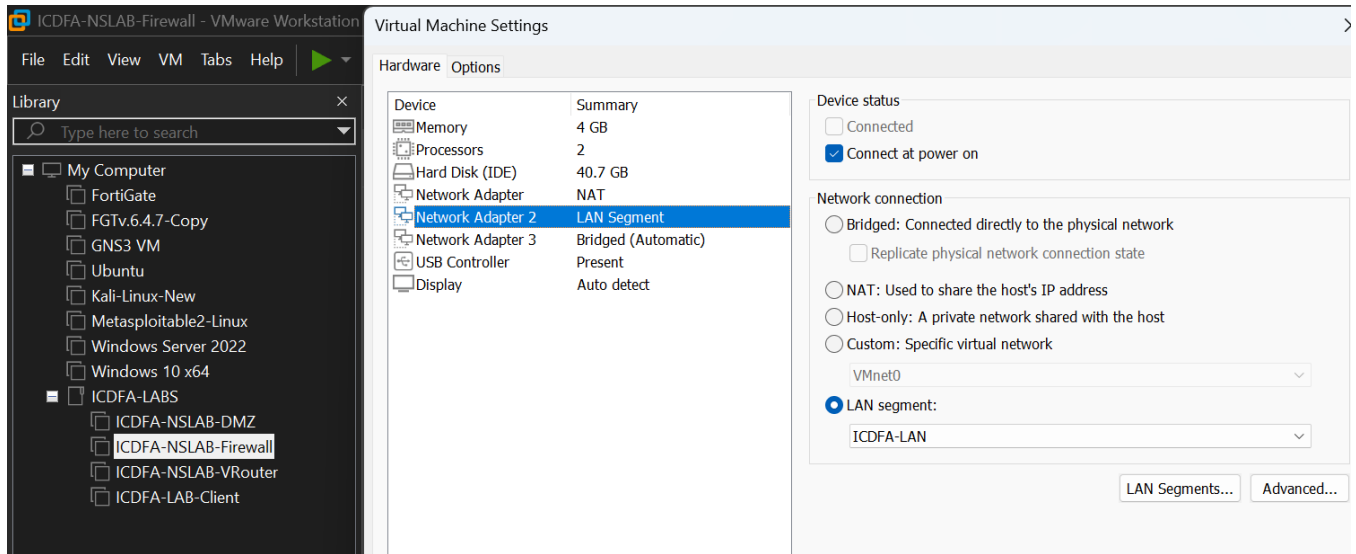
The network for this lab consists of one firewall (OPNSense) and an Ubuntu client behind the Firewall. The lab focuses on commissioning OPNSense Firewall and Ubuntu virtual machines in VMware and verifying connectivity between the virtual machines and the internet.

Part A: Prepare VMware

The attached screenshot shows the virtual machine settings for the Ubuntu client (ICDFA-LAB-Client). Its network adapter was set to the ICDFA-LAN , LAN Segment.



The screenshot below shows the VM settings for the Firewall (ICDFA-NSLAB-Firewall). The first network adapter is connected to NAT, the second adapter is connected to the LAN segment (ICDFA-LAN). The third adapter was left at the default setting as it was not used in the lab.



Part B: Verify OPNsense interfaces

The OPNsense firewall has 3 interfaces. The interfaces used in the lab are the WAN and LAN. The WAN interface is set to receive an IP address through DHCP. The LAN interface has the IP address 10.10.10.1/24 and is set to give out IP addresses to clients in the LAN through DHCP.

```
*** icdfa-nsrab-firewall-v1.icdfa.test: OPNsense 26.7 (amd64) ***

DMZ (em2)      -> v4: 10.10.20.1/24
LAN (em1)      -> v4: 10.10.10.1/24
WAN (em0)      -> v4/DHCP4: 192.168.219.165/24

HTTPS: SHA256 E8 0B 30 3E D5 89 EB 63 A0 D8 83 2F E6 38 4C 22
          82 2C 5A A4 39 D5 45 54 57 50 A0 73 58 E0 46 90
SSH:      SHA256 K10LtdmW484dUmuPy37tD714RA0De5LKjSU/+bKB1Tw (ECDSA)
SSH:      SHA256 5wvyP8AD885sP0fHz7CgfNStFMhPwXX42x23cBxco4 (ED25519)
SSH:      SHA256 6e/0XrjSdfL/Hsy8at2w0IPQGsblP6zFfuJ/F250d+A (RSA)

0) Logout
1) Assign interfaces
2) Set interface IP address
3) Reset the root password
4) Reset to factory defaults
5) Power off system
6) Reboot system

7) Ping host
8) Shell
9) pfTop
10) Firewall log
11) Reload all services
12) Update from console
13) Restore a backup

Enter an option: █
```

Part C Verify Ubuntu addressing

The Ubuntu client on the same LAN segment as the LAN interface of the firewall, received the IP address 10.10.10.54. The attached image shows the IP address and route for the client.

```
icdfa@icdfa-nslab-client-v1:~$ ip -4 -br address
lo                UNKNOWN    127.0.0.1/8
ens33             UP          10.10.10.54/24
icdfa@icdfa-nslab-client-v1:~$ ip route
default via 10.10.10.1 dev ens33 proto dhcp src 10.10.10.54 metric 100
10.10.10.0/24 dev ens33 proto kernel scope link src 10.10.10.54 metric 100
icdfa@icdfa-nslab-client-v1:~$
```

Part D Perform connectivity tests

Next, tests were run to confirm client connectivity. The screenshots below shows a progressive ping test.

Results from ping test to LAN interface IP address (10.10.10.1) and DNS server (1.1.1.1) to confirm that routing and outbound NAT are working

```
icdfa@icdfa-nslab-client-v1:~$ ping -c 5 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=1.09 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=3.99 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=0.460 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=0.369 ms
64 bytes from 10.10.10.1: icmp_seq=5 ttl=64 time=0.457 ms

--- 10.10.10.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4046ms
rtt min/avg/max/mdev = 0.369/1.272/3.985/1.380 ms
icdfa@icdfa-nslab-client-v1:~$ ping -c 5 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=127 time=8.93 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=127 time=6.05 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=127 time=7.29 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=127 time=7.46 ms
64 bytes from 1.1.1.1: icmp_seq=5 ttl=127 time=7.03 ms

--- 1.1.1.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 6.049/7.350/8.926/0.927 ms
```

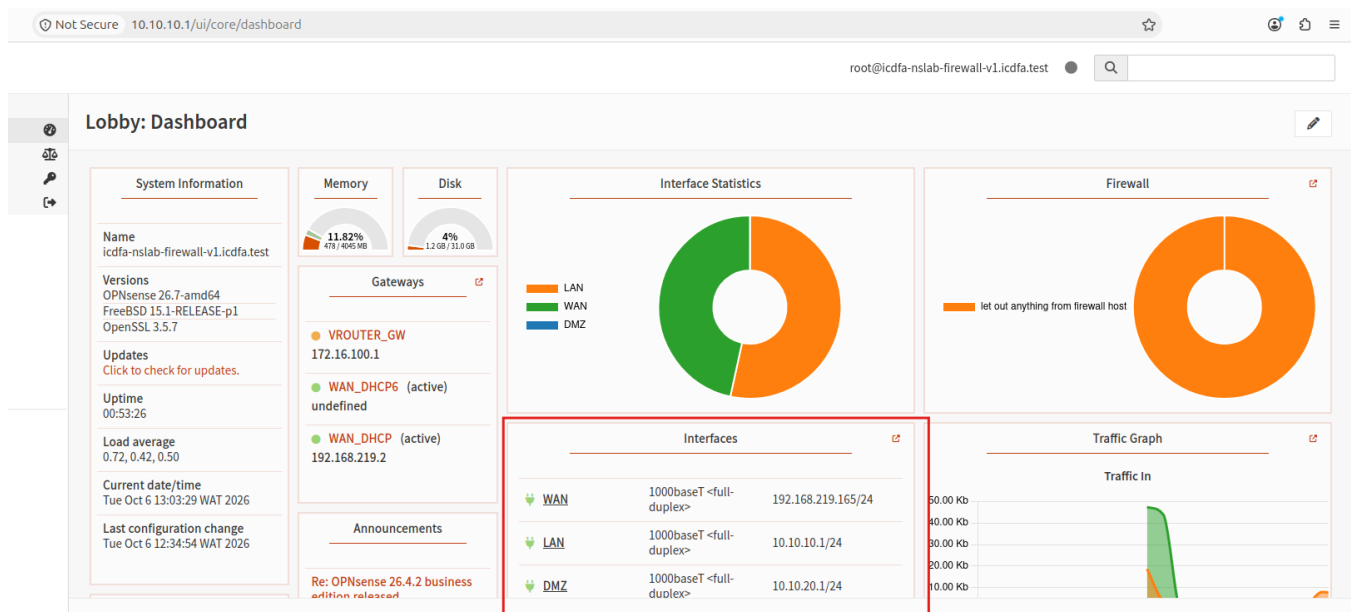
Results from the getent hosts command to verify proper configuration of and communication with a DNS server.

```
icdfa@icdfa-nslab-client-v1:~$ getent hosts opnsense.org
2001:1af8:2050:a001:1::1 opnsense.org
icdfa@icdfa-nslab-client-v1:~$
```

Results to verify HTTP/HTTPS connectivity.

```
icdfa@icdfa-nsfab-client-v1:~$ curl -I https://opnsense.org
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 11:59:26 GMT
Server: Apache
Link: <https://opnsense.org/wp-json/>; rel="https://api.w.org/", <https://opnsense.org/wp-json/wp/v2/pages/199>; rel="alternate"; title="JSON"; type="application/json", <https://opnsense.org/>; rel=shortlink
Strict-Transport-Security: max-age=15552000; includeSubDomains; preload
Content-Type: text/html; charset=UTF-8
```

The image below shows the OPNSense dashboard visible after logging into the LAN interface address on the client device.



Part E: Observe packets in Wireshark

The images below show the results of filtering for specific protocols after a packet capture was performed on the Ubuntu VM.

ICMP was used to identify the echo request and echo reply packets.

No.	Time	Source	Destination	Protocol	Length	Info
151	44.372476644	10.10.10.54	10.10.10.1	ICMP	98	Echo (ping) request id=0x1932, seq=1/256, ttl=64 (reply in 152)
152	44.372837362	10.10.10.1	10.10.10.54	ICMP	98	Echo (ping) reply id=0x1932, seq=1/256, ttl=64 (request in 151)
153	45.415156625	10.10.10.54	10.10.10.1	ICMP	98	Echo (ping) request id=0x1932, seq=2/512, ttl=64 (reply in 154)
154	45.415520442	10.10.10.1	10.10.10.54	ICMP	98	Echo (ping) reply id=0x1932, seq=2/512, ttl=64 (request in 153)
169	46.439228199	10.10.10.54	10.10.10.1	ICMP	98	Echo (ping) request id=0x1932, seq=3/768, ttl=64 (reply in 170)
170	46.439658629	10.10.10.1	10.10.10.54	ICMP	98	Echo (ping) reply id=0x1932, seq=3/768, ttl=64 (request in 169)
171	47.462818708	10.10.10.54	10.10.10.1	ICMP	98	Echo (ping) request id=0x1932, seq=4/1024, ttl=64 (reply in 172)
172	47.463229636	10.10.10.1	10.10.10.54	ICMP	98	Echo (ping) reply id=0x1932, seq=4/1024, ttl=64 (request in 171)
181	48.487203493	10.10.10.54	10.10.10.1	ICMP	98	Echo (ping) request id=0x1932, seq=5/1280, ttl=64 (reply in 182)
182	48.487753752	10.10.10.1	10.10.10.54	ICMP	98	Echo (ping) reply id=0x1932, seq=5/1280, ttl=64 (request in 181)
195	52.075647432	10.10.10.54	10.10.10.1	ICMP	98	Echo (ping) request id=0x1935, seq=1/256, ttl=64 (reply in 196)
196	52.076106199	10.10.10.1	10.10.10.54	ICMP	98	Echo (ping) reply id=0x1935, seq=1/256, ttl=64 (request in 195)
197	53.095190687	10.10.10.54	10.10.10.1	ICMP	98	Echo (ping) request id=0x1935, seq=2/512, ttl=64 (reply in 198)
198	53.096612513	10.10.10.1	10.10.10.54	ICMP	98	Echo (ping) reply id=0x1935, seq=2/512, ttl=64 (request in 197)
199	54.097065116	10.10.10.54	10.10.10.1	ICMP	98	Echo (ping) request id=0x1935, seq=3/768, ttl=64 (reply in 200)
200	54.097543304	10.10.10.1	10.10.10.54	ICMP	98	Echo (ping) reply id=0x1935, seq=3/768, ttl=64 (request in 199)

DNS was used to identify the DNS query for opnsense.org

No.	Time	Source	Destination	Protocol	Length	Info
98	19.028035640	10.10.10.54	10.10.10.1	DNS	101	Standard query 0x9dfc A incoming.telemetry.mozilla.org OPT
99	19.028412639	10.10.10.54	10.10.10.1	DNS	101	Standard query 0xad0f AAAA incoming.telemetry.mozilla.org OPT
134	19.785767103	10.10.10.1	10.10.10.54	DNS	238	Standard query response 0x9dfc A incoming.telemetry.mozilla.org CNAME telemetry-incomin...
135	20.171560886	10.10.10.1	10.10.10.54	DNS	312	Standard query response 0xad0f AAAA incoming.telemetry.mozilla.org CNAME telemetry-inco...
136	20.172969935	10.10.10.54	10.10.10.1	DNS	119	Standard query 0x21c5 AAAA prod.ingestion-edge.prod.dataservices.mozgcp.net OPT
137	20.173330222	10.10.10.1	10.10.10.54	DNS	212	Standard query response 0x21c5 AAAA prod.ingestion-edge.prod.dataservices.mozgcp.net SO...
164	20.821840166	10.10.10.54	10.10.10.1	DNS	119	Standard query 0xae69 AAAA prod.ingestion-edge.prod.dataservices.mozgcp.net OPT
165	20.822507274	10.10.10.1	10.10.10.54	DNS	212	Standard query response 0xae69 AAAA prod.ingestion-edge.prod.dataservices.mozgcp.net SO...
196	21.455545279	10.10.10.54	10.10.10.1	DNS	119	Standard query 0xc91b AAAA prod.ingestion-edge.prod.dataservices.mozgcp.net OPT
197	21.456141098	10.10.10.1	10.10.10.54	DNS	212	Standard query response 0xc91b AAAA prod.ingestion-edge.prod.dataservices.mozgcp.net SO...

ARP was used to identify the MAC address for the LAN interface with IP address 10.10.10.1

No.	Time	Source	Destination	Protocol	Length	Info
288	347.417697830	VMware_03:b3:78	VMware_0f:34:28	ARP	42	Who has 10.10.10.1? Tell 10.10.10.54
289	347.418354177	VMware_0f:34:28	VMware_03:b3:78	ARP	60	10.10.10.1 is at 00:0c:29:0f:34:28

Default Gateway Explanation

The Ubuntu Client uses 10.10.10.1 as its default gateway because that address belongs to the firewall's LAN interface. The firewall's LAN interface is the only path the client has to reach anything outside its subnet (10.10.10.0/24).

Completion questions

- What is the difference between the OPNsense WAN and LAN interfaces?
 LAN interface is the interface that acts as a gateway for internal clients. It is usually a static IP address.
 WAN interface is the outside-facing interface that connects to the internet. It usually gets its address through DHCP.
- Why must both internal adapters use the same VirtualBox network name?
 The LAN and network adapters for the firewall and Ubuntu client must have the same name so that they can be on the same LAN segment. This allows for frames to pass between them, allowing communication.
- What information does the default route provide to Ubuntu?
 The default route tells ubuntu where to send packets that have a destination address outside its subnet.
 The default route here is 10.10.10.1
- Which packet exchange allows Ubuntu to learn the firewall MAC address?
 An ARP request allows Ubuntu to send a broadcast message to find the firewall MAC address. The Firewall returns an ARP reply containing its MAC address.
- Why does a successful ping to 1.1.1.1 not automatically prove that DNS is working?

Ping uses the IP address, so no domain name lookup occurs. Hence a successful ping to 1.1.1.1 proves only that routing and NAT and not that DNS is working.